

Rigorous Generalization of a Double Summation Limit

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Abstract

This article generalizes the double limit summation presented in the reference file

`dailyintegral-limit-limit-1x1-2026-08-17.jpg`. We construct a parametrized family of sums, rigorously prove convergence using Tannery's Theorem (a discrete analog of the Dominated Convergence Theorem), and express the closed-form limit in terms of the Riemann zeta function.

1 Introduction

The problem of interest, as depicted in the file `dailyintegral-limit-limit-1x1-2026-08-17.jpg`, asks to evaluate the following limit:

$$L = \lim_{n \rightarrow \infty} \sum_{k=1}^n \sum_{j=1}^n \frac{2k^{-\frac{2j}{n}} \ln(k)}{nk^2} \quad (1)$$

We can generalize this limit by replacing the constants with positive real parameters p and q . Let us define the generalized sequence of sums $S_n(p, q)$ for parameters $p > 0$ and $q > 1$:

$$S_n(p, q) = \sum_{k=1}^n \sum_{j=1}^n \frac{pk^{-\frac{pj}{n}} \ln(k)}{nk^q} \quad (2)$$

The goal is to rigorously evaluate $\lim_{n \rightarrow \infty} S_n(p, q)$.

2 Mathematical Formulation

To separate the variables, we rewrite the double summation by grouping terms dependent on the inner index j :

$$S_n(p, q) = \sum_{k=1}^n \frac{p \ln(k)}{k^q} \left(\frac{1}{n} \sum_{j=1}^n k^{-\frac{pj}{n}} \right) \quad (3)$$

Let us define the inner term $a_k(n)$ as:

$$a_k(n) = \frac{p \ln(k)}{k^q} \left(\frac{1}{n} \sum_{j=1}^n (k^{-p})^{\frac{j}{n}} \right) \quad (4)$$

Thus, our limit is $\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k(n)$. Because both the number of terms and the terms themselves depend on n , we cannot blindly interchange the limit and the summation. We must employ Tannery's Theorem.

Lemma 1 (Tannery's Theorem). *Let $a_k(n)$ be a sequence of terms such that $\lim_{n \rightarrow \infty} a_k(n) = a_k$ for each fixed k . If there exists a bounding sequence M_k independent of n such that $|a_k(n)| \leq M_k$ for all n and k , and if the infinite series $\sum_{k=1}^{\infty} M_k$ converges, then:*

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k(n) = \sum_{k=1}^{\infty} \lim_{n \rightarrow \infty} a_k(n) = \sum_{k=1}^{\infty} a_k$$

3 Rigorous Evaluation

Theorem 1. *For any real numbers $p > 0$ and $q > 1$, the generalized limit evaluates to:*

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \sum_{j=1}^n \frac{pk^{-\frac{pj}{n}} \ln(k)}{nk^q} = \zeta(q) - \zeta(p+q)$$

where $\zeta(s)$ is the Riemann zeta function.

Proof. First, we find the pointwise limit $a_k = \lim_{n \rightarrow \infty} a_k(n)$ for a fixed k . For $k = 1$, $a_1(n) = 0$ for all n , so $a_1 = 0$. For fixed $k \geq 2$, the expression in the parenthesis is a right Riemann sum for the integral of the function $f(x) = k^{-px}$ over the interval $[0, 1]$:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n (k^{-p})^{\frac{j}{n}} = \int_0^1 k^{-px} dx = \left[\frac{k^{-px}}{-p \ln(k)} \right]_0^1 = \frac{1 - k^{-p}}{p \ln(k)} \quad (5)$$

Therefore, the pointwise limit for $k \geq 2$ is:

$$a_k = \lim_{n \rightarrow \infty} a_k(n) = \frac{p \ln(k)}{k^q} \left(\frac{1 - k^{-p}}{p \ln(k)} \right) = \frac{1 - k^{-p}}{k^q} = \frac{1}{k^q} - \frac{1}{k^{p+q}} \quad (6)$$

Notice that for $k = 1$, the formula gives $\frac{1}{1^q} - \frac{1}{1^{p+q}} = 0$, which matches our direct observation.

Second, we must find an absolutely convergent bounding sequence M_k . For $k \geq 2$ and $p > 0$, the function $f(x) = k^{-px}$ is strictly decreasing on $[0, 1]$. A fundamental property of Riemann sums for decreasing functions dictates that the right Riemann sum is strictly bounded above by the definite integral:

$$\frac{1}{n} \sum_{j=1}^n k^{-\frac{pj}{n}} \leq \int_0^1 k^{-px} dx = \frac{1 - k^{-p}}{p \ln(k)} \quad (7)$$

Substituting this inequality back into our expression for $a_k(n)$:

$$|a_k(n)| \leq \frac{p \ln(k)}{k^q} \left(\frac{1 - k^{-p}}{p \ln(k)} \right) = \frac{1}{k^q} - \frac{1}{k^{p+q}} < \frac{1}{k^q} \quad (8)$$

Let $M_k = \frac{1}{k^q}$ for $k \geq 1$. Since we established $q > 1$, the sum $\sum_{k=1}^{\infty} M_k = \sum_{k=1}^{\infty} \frac{1}{k^q} = \zeta(q)$ converges.

By Tannery's Theorem, the conditions are satisfied to interchange the limit and the sum:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k(n) = \sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} \left(\frac{1}{k^q} - \frac{1}{k^{p+q}} \right) \quad (9)$$

By separating the absolutely convergent series (since $q > 1$ and $p + q > 1$), we achieve our generalized result:

$$\sum_{k=1}^{\infty} \frac{1}{k^q} - \sum_{k=1}^{\infty} \frac{1}{k^{p+q}} = \zeta(q) - \zeta(p+q) \quad (10)$$

□

4 Conclusion and Specific Application

Corollary 1. *The specific limit shown in the file `dailyintegral-limit-limit-1x1-2026-08-17.jpg` converges to $\frac{\pi^2}{6} - \frac{\pi^4}{90}$.*

Proof. Applying Theorem 1 with parameters $p = 2$ and $q = 2$, the limit is precisely evaluated as:

$$L = \zeta(2) - \zeta(4) \quad (11)$$

Using the well-known values evaluated by Euler, $\zeta(2) = \frac{\pi^2}{6}$ and $\zeta(4) = \frac{\pi^4}{90}$, we conclude:

$$L = \frac{\pi^2}{6} - \frac{\pi^4}{90} \approx 0.56157 \quad (12)$$

□